Sheng Kuang

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Education

Shihezi University Sept. 2024 – Expected Jun. 2028
B.S. in Biological Science; Weighted Average: 85.1/100, Rank: 27/121 *Shihezi, Xinjiang, China*

Relevant Coursework: Ecology (91.2), Life Science Instruments and Technology (89.7), Plant Morphology and Anatomy (87.6), Biochemistry Experiment (96.3), Molecular Biology (85.1)

Honors: Third-class Scholarship (2025, 2026); Xinjiang TaiKun Group Enterprise Scholarship (2025)

Research Experience

Multi-omics Analysis of Transposable Elements and Their Roles in Salt and Drought Stress Responses in *Glycyrrhiza* Jan. 2026 – Present
Team Leader *Shihezi University, China*

- Led a multi-omics investigation of transposable element (TE) dynamics in three *Glycyrrhiza* species under salt and drought stress, integrating **458,673 TE annotations**, **99 RNA-seq libraries**, and **189 whole-genome resequencing datasets**.
- Developed a custom **Bash pipeline** integrating six structural signals for non-reference TE insertion detection across the **189** resequenced genomes, and subsequently validated it against simulated ground truth — a process that exposed structural errors in the original detection logic, which I then corrected.
- Integrated RNA-seq, WGCNA, and machine learning to investigate TE-associated stress responses; developed an interpretable **XGBoost-SHAP** framework with 5-fold cross-validation (**AUC = 0.942**) to prioritize stress-associated TE candidates.

Genome-Wide Identification of the IFS Gene Family and ARF-Mediated Regulation Mar. 2025 – Aug. 2025
Undergraduate Researcher, Prof. Shandang Shi & Prof. Shuangquan Xie *Shihezi University, China*

- Conducted genome-wide identification of the isoflavone synthase (IFS) gene family across three *Glycyrrhiza* species, identifying **28 IFS genes** by BLAST homology search and Pfam/NCBI-CDD verification, and reconstructed their phylogeny (MEGA X, 1,000 bootstrap replicates), gene structure, conserved motifs and synteny.
- Analysed promoter regions with PlantCARE to identify hormone- and stress-responsive cis-elements, and integrated the computational work with wet-lab validation; co-authored a paper in *International Journal of Molecular Sciences* (2025, 26, 10435).

Protein Design Track, 5th SynBio Challenges — GFP Variant Design Jan. 2026 – Aug. 2026
Team Leader *Shenzhen, China*

- Led a team in the Facility Track — Protein Design: we designed GFP variants optimised for fluorescence brightness and thermostability, and submitted six predicted mutants for experimental testing.
- Used **AlphaFold** and **Rosetta** to generate and rank variants; the sequences were synthesised and characterised by the Shenzhen synthetic biology facility (cell-free expression, initial and post-heat fluorescence).

Student Achievement Visualization Teaching Evaluation Platform Feb. 2026 – Aug. 2026
Co-Principal Investigator (Second Leader), supervised by Prof. Xia Bo & Prof. Xu Liming *Shihezi University, China*

- Co-led development of a full-stack educational assessment platform, designing PostgreSQL schemas, RESTful APIs (Node.js/Express) and role-based access control with tag-level data isolation.
- Built ECharts competency visualisations and batch PDF reporting, and containerised the platform with Docker Compose for pilot deployment.

Publications

Xu, Q.; et al. (incl. **Kuang, S.**). Genome-Wide Identification and Characterization of Isoflavone Synthase (IFS) Gene Family, and Analysis of GgARF4-GgIFS9 Regulatory Module in *Glycyrrhiza glabra*. *Int. J. Mol. Sci.* **2025**, *26*, 10435. DOI: 10.3390/ijms262110435.

Honors & Awards

2026 **Silver Award** (team leader), 5th SynBio Challenges, Shenzhen
2026 **National Third Prize**, Chinese Collegiate Computing Competition (4C)
2026 **Third Prize** (Xinjiang), National Undergraduate Life Science Competition; **Third Prize** (Xinjiang), NECCS
2025 **First Prize**, National Cotton and Hemp Innovation and Entrepreneurship Competition
2025 **Outstanding Student Cadre**, Shihezi University

Technical Skills

Bioinformatics RNA-seq analysis (HISAT2, featureCounts, DESeq2); TE annotation; phylogenetic analysis (IQ-TREE, MrBayes, RAxML-NG); synteny and genomic analysis; WGCNA; BEDTools
Programming R, Python, Bash/Shell; C/C++; Java; HTML/CSS, JavaScript; Vue3, Node.js/Express, PostgreSQL, Docker
Machine Learning XGBoost, SHAP, cross-validation, and model evaluation
Molecular Biology DNA/RNA extraction, PCR, molecular cloning